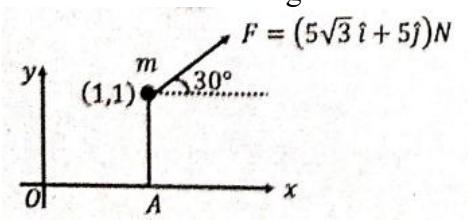
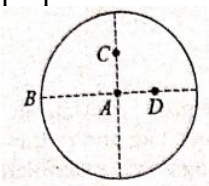
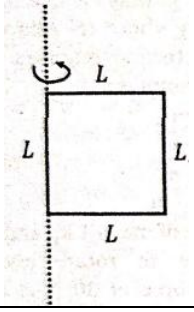
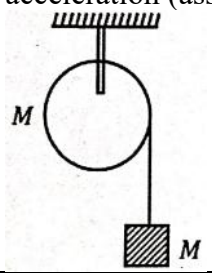
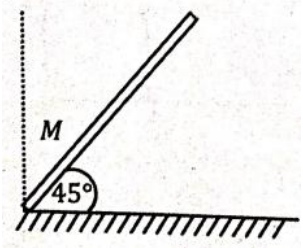
Question	A forcer $\vec{F}=\hat{i}+2\hat{j}+3\hat{k}N$ acts a points $4\hat{i}+2\hat{j}-\hat{k}m$ The magnitude of the torque about the point $\left(\hat{i}+2\hat{j}+\hat{k}\right)m$ is $\sqrt{x}N-m.$ The value of x is
Type	multiple choice
Option	145
Option	195
Option	245
Option	295
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	The moment of force applied on a door is 18 N-m and a force 4.5 N is applied by a boy perpendicular to plane of door, the distance of the handle from the pivot is
Type	multiple choice
Option	0.25 m
Option	0.75 m
Option	2.25 m
Option	4.0 m
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	A couple acting on a body produces:
Type	multiple choice
Option	Rotation without translation
Option	Translation without rotation
Option	Both rotation and translation
Option	Can't say
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	The torque about the point 0(3, -1) due to a force $\vec{F}(2,4)N$ on a plane $X-Y$ passing through the point $A(5,2)$ is:
Type	multiple choice
Option	$2\hat{k}\,N.m$
Option	$-2\hat{k}\,N.m$
Option	$14\hat{k}\,N.m$
Option	$-14\hat{k}\,N.m$
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	A force $-F\hat{k}$ acts on 0, the origin of the coordinate system. The toque about the point $(1,-1)$ will be:
Type	multiple choice
Option	$-F\left(\hat{i}+\hat{j}\right)$

Option	$F\left(\hat{i}+\hat{j}\right)$
Option	$-F\left(\hat{i}-\hat{j}\right)$
Option	$F\left(\hat{i}-\hat{j}\right)$
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	Which of the following is the value of the torque of force F about origin 0 
Type	multiple choice
Option	$\vec{\tau} = 5(1-\sqrt{3})kN.m$
Option	$\vec{\tau} = 5(1-\sqrt{3})\hat{J}N.m$
Option	$\vec{\tau} = 5(\sqrt{3}+1)\hat{i}N.m$
Option	$\vec{\tau} = \sqrt{3}\,jN.m$
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	If $\vec{F}$ is the force acting on a particle having position vector $\vec{r}$ and $\vec{\tau}$ be the torque of this force about the origin, then.
Type	multiple choice
Option	$\vec{r}.\vec{\tau} \neq 0$ and $\vec{F}.\vec{\tau} = 0$
Option	$\vec{r}.\vec{\tau} > 0$ and $\vec{F}.\vec{\tau} < 0$
Option	$\vec{r}.\vec{\tau} = 0$ and $\vec{F}.\vec{\tau} = 0$
Option	$\vec{r}.\vec{\tau} \neq 0$ and $\vec{F}.\vec{\tau} \neq 0$
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	The moment of the force, $\vec{F} = 4\hat{i} + 5\hat{j} - 6\hat{k}$ at point $(2,0,-3)$ about the point $(2,-2,-2)$ is given by:
Type	multiple choice
Option	$-8\hat{i} - 4\hat{j} - 7\hat{k}$
Option	$-7\hat{i} - 8\hat{j} - 4\hat{k}$
Option	$-7\hat{i} - 4\hat{j} - 8\hat{k}$
Option	$-6\hat{i} - 4\hat{j} - 8\hat{k}$
Answer	3
Solution	
Positive Marks	4
Negative Marks	1

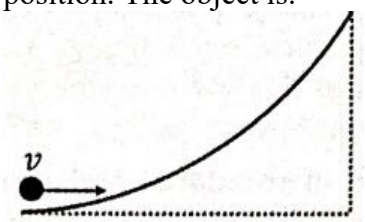
Question	Five particles of mass 2 kg each are attached to the circumference of a circular disc of a radius of 0.1 m and negligible mass. The moment of inertia of the system about the axis passing through the centre of the disc and perpendicular to its plane will be:
Type	multiple choice
Option	$1\ kg-m^2$
Option	$0.2\ kg-m^2$
Option	$2\ kg-m^2$
Option	$0.1\ kg-m^2$
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	The moment of inertia of a horizontal ring about its vertical axis through the centre is mR^2. The moment of inertia about its tangent parallel to the plane is:
Type	multiple choice
Option	$\frac{3mR^2}{2}$
Option	$\frac{mR^2}{4}$
Option	$\frac{mR^2}{2}$
Option	$\frac{3mR^2}{4}$
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	The moment of inertia of a uniform circular disc of radius 'R' and mass 'M' about an axis touching the end of its diameter and normal to the disc will be:
Type	multiple choice
Option	$\frac{3}{2}MR^2$
Option	$\frac{1}{2}MR^2$
Option	MR^2
Option	$\frac{2}{5}MR^2$
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	The moment of inertia of a uniform circular disc is maximum about an axis perpendicular to the disc and passing through: 

Type	multiple choice
Option	B
Option	C
Option	D
Option	A
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	Four thin rods, each of mass m and length L, form a square. The moment of inertia of system about given axis is 
Type	multiple choice
Option	$\frac{5}{3}mL^2$
Option	$4mL^2$
Option	$\frac{mL^2}{4}$
Option	$\frac{2}{3}mL^2$
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	Two uniform, thin, identical rods, each of mass M and length l are joined together to form a cross. What will be the moment of inertia of the cross about a bisector in the plane of rod?
Type	multiple choice
Option	$\frac{Ml^2}{6}$
Option	$\frac{Ml^2}{8}$
Option	$\frac{Ml^2}{12}$
Option	$\frac{Ml^2}{3}$
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	The moment of inertia of a thin uniform circular disc about one of its diameter is I. Its moment of inertia about an axis perpendicular to the circular surface and passing through its center will be:

Type	multiple choice
Option	$\sqrt{2}I$
Option	$3I$
Option	$2I$
Option	$\frac{I}{\sqrt{2}}$
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	A constant torque of 1000 N-m turns wheel of moment of inertial $200\,kg-m^2$ about an axis through its centre from rest. Its angular velocity after 3 s is.
Type	multiple choice
Option	$1\,rad/s$
Option	$5\,rad/s$
Option	$10\,rad/s$
Option	$15\,rad/s$
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	An automobile engine develops 100 kW when rotating at a speed of 1800 rev/min. What torque does it deliver?
Type	multiple choice
Option	350 N-m
Option	440 N-m
Option	531 N-m
Option	628 N-m
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	A mass M is supported by a massless string wound round a uniform cylinder of mass M and radius R. On releasing the mass from rest, it will fall with an acceleration (assuming no slipping). 
Type	multiple choice
Option	$\frac{g}{3}$
Option	$\frac{g}{2}$
Option	$\frac{2g}{3}$
Option	g

Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	A rod released at an angle 45^0 to the horizontal as shown. If its length is L, its angular acceleration about its foot at ground is initially. 
Type	multiple choice
Option	$\frac{3g}{2L}$
Option	$\frac{3g}{\sqrt{2}L}$
Option	$\frac{3g}{2\sqrt{2}L}$
Option	$\frac{6g}{\sqrt{2}L}$
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	A rope is wound around a hollow cylinder of mass 3kg and radius 40 cm. What is the angular acceleration of the cylinder, if the rope is pulled with a force of 30N?
Type	multiple choice
Option	$25 m/s^2$
Option	$0.25 rad/s^2$
Option	$25 rad/s^2$
Option	$5 m/s^2$
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	A solid cylinder of mass 2 kg and radius 4 cm is rotating about its axis the rate of 3 rpm. The torque required to stop after
Type	multiple choice
Option	$2\times10^6 N-m$
Option	$2\times10^{-6} N-m$
Option	$2\times10^{-3} N-m$
Option	$12\times10^{-4} N-m$
Answer	2
Solution	
Positive Marks	4

Negative Marks	1
Question	A hollow sphere of mass 1 kg and radius 10 cm is free to rotate about its diameter. If a force of 30 N is applied tangentially to it, its angular acceleration is (in rad/s^2).
Type	multiple choice
Option	5000
Option	450
Option	50
Option	5
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	A wheel having moment of inertia $4kg\,m^2$ about its symmetrical axis, rotates at rate of 240 rpm about it. The torque which can stop the rotation of the wheel in one minute is.
Type	multiple choice
Option	$\frac{7\pi}{15}Nm$
Option	$\frac{8\pi}{15}Nm$
Option	$\frac{2\pi}{9}Nm$
Option	$\frac{3\pi}{7}Nm$
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	A coin of mass 10 g rolls along a horizontal table with a velocity of 6 cm/s. Its total kinetic energy is
Type	multiple choice
Option	$9\mu J$
Option	$18\mu J$
Option	$27\mu J$
Option	$36\mu J$
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	A solid sphere is in rolling motion. In rolling motion a body possesses translational kinetic energy (K_t) as well as rotational kinetic energy (K_r) simultaneously. The ratio $k_t:(k_t+k_r)$ for the sphere is
Type	multiple choice
Option	$7:10$
Option	$5:7$
Option	$10:7$
Option	$2:5$
Answer	2

Solution	
Positive Marks	4
Negative Marks	1
Question	A solid sphere of mass 1kg, radius 10 cm rolls down an inclined plane of height 7m. The velocity of its centre as it reaches the ground level is.
Type	multiple choice
Option	$7\,m/s$
Option	$10\,m/s$
Option	$15\,m/s$
Option	$20\,m/s$
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	Two solid disc of radii r and 2r roll from the top of an inclined plane without slipping. Then
Type	multiple choice
Option	The bigger disc will reach the horizontal level first
Option	The smaller disc will reach the horizontal level first
Option	The time difference of reaching of the discs at the horizontal level will depend on the inclination of the plane
Option	Both the discs will reach at the same time
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	A small object of uniform density rolls up a curved surface with an initial velocity v. It reaches up to a maximum height of $\frac{3v^2}{4g}$ with respect to the initial position. The object is. 
Type	multiple choice
Option	Ring
Option	Solid sphere
Option	Holow sphere
Option	Disc
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	A body is rolling down an incline plane. Its translational and rotational kinetic energies are equal. The body is a
Type	multiple choice
Option	Solid sphere
Option	Hollow sphere
Option	Solid cylinder
Option	Hollow cylinder

Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	A solid cylinder rolls without slipping down an inclined plane at an angle 30° with the horizontal. The acceleration of the cylinder is.
Type	multiple choice
Option	$\frac{g}{2}$
Option	$\frac{g}{3}$
Option	g
Option	$\frac{2}{3}g$
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	A hollow sphere of diameter 0.2 m and mass 2 kg is rolling on an inclined plane with velocity v = 0.5 m/s. The kinetic energy of the sphere is
Type	multiple choice
Option	0.1 J
Option	0.3 J
Option	0.5 J
Option	0.42 J
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	The ratio of rotational and translatory kinetic energies of a solid sphere which is rolling without slipping on a horizontal plane is
Type	multiple choice
Option	$2/9$
Option	$2/7$
Option	$2/5$
Option	$7/2$
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	When a sphere of moment of inertia I about its centre of gravity and mass 'm' rolls from rest down an inclined plane without slipping, its kinetic energy is
Type	multiple choice
Option	$\frac{1}{2}I\omega^2$
Option	$\frac{1}{2}mv^2$
Option	$I\omega + mv$
Option	$\frac{1}{2}I\omega^2 + \frac{1}{2}mv^2$
Answer	4

Solution	
Positive Marks	4
Negative Marks	1
Question	If rotational kinetic energy is 50% of translational kinetic energy, then the body is
Type	multiple choice
Option	Ring
Option	Solid Cylinder
Option	Hollow sphere
Option	Solid sphere
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	The M.I. of a body about the given axis is 1.2 $kg\,m^2$, initially the body is at rest. In order to produce a rotational kinetic energy of 1500 J, an angular acceleration of $25\ rad/\sec^2$ must be applied about that axis for duration of
Type	multiple choice
Option	4 sec
Option	2 sec
Option	8 sec
Option	10 sec
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	The total kinetic energy of a body of mass 10 kg and radius 0.5 m moving with a velocity of 2 m/s without slipping is 32.8 joule. The radius of gyration of the body is
Type	multiple choice
Option	0.25 m
Option	0.2 m
Option	0.5 m
Option	0.4 m
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	A disc of moment of inertia $\frac{9.8}{\pi^2} kg\ m^2$ is rotating at 600 rpm. If the frequency of rotation changes from 600 rpm to 300 rpm, then what is the magnitude of work done
Type	multiple choice
Option	1470 J
Option	1452 J
Option	1567 J
Option	1632 J
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	A ball rolls without slipping. The radius of gyration of the ball about an axis

	passing through its centre of mass is K. If radius of the ball be R, then the fraction of total energy associated with its rotational energy will be
Type	multiple choice
Option	$\frac{K^2}{R^2}$
Option	$\frac{K^2}{K^2+R^2}$
Option	$\frac{R^2}{K^2+R^2}$
Option	$\frac{K^2+R^2}{R^2}$
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	Ratio of total kinetic energy and rotational kinetic energy in the motion of a disc is
Type	multiple choice
Option	$1:1$
Option	$7:2$
Option	$2:1$
Option	$3:1$
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	A solid sphere of mass m and radius R is rotating about its diameter. A solid cylinder of the same mass and same radius is also rotating about its geometrical axis with an angular speed twice that of the sphere. The ratio of their kinetic energies of rotation $E_{sphere} / E_{cylinder}$ will be
Type	multiple choice
Option	31
Option	23
Option	15
Option	14
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	A particle is moving in a circular path. The correct statement out of the following is
Type	multiple choice
Option	Angular momentum will be conserved but linear momentum will not be conserved
Option	Angular momentum will not be conserved
Option	Linear momentum will be conserved
Option	Both will be conserved
Answer	1
Solution	
Positive Marks	4
Negative Marks	1

Question	A solid sphere is rotating about a diameter at an angular velocity ω. If it cools so that its radius reduces to $\frac{1}{n}$ of its original value, its angular velocity becomes
Type	multiple choice
Option	$\frac{\omega}{n}$
Option	$\frac{\omega}{n^2}$
Option	$n\omega$
Option	$n^2\omega$
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	Two bodies have their moments of inertia I and $2I$ respectively about their axis of rotation. If their kinetic energies of rotation are equal, their angular momentum will be in the ratio
Type	multiple choice
Option	12
Option	$\sqrt{2}:1$
Option	2 1
Option	$1:\sqrt{2}$
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	A man stands on a turn-table stretching with two equal loads in hand and rotates. Then he folds his arm. Which of the following statement is correct
Type	multiple choice
Option	Linear momentum is conserved
Option	Kinetic energy decreases
Option	Angular momentum increases
Option	Angular velocity increases
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	The angular momentum of a system of particles is not conserved
Type	multiple choice
Option	When a net external force acts upon the system
Option	When a net external torque is acting upon the system
Option	When a net external impulse is acting upon the system
Option	None of these
Answer	2
Solution	
Positive Marks	4
Negative Marks	1